Lesson plan · for the teacher

What makes a choice the best?

Optimization · Level 1 · Lesson 1

Time	Materials
40 min	Board and chalk; Five cards per group, each with an outing (or a class reward), its cost and its votes; A budget on the board, such as 1,200 coins

Learning goals

- Name the options, the limits and the goal of a decision.
- Tell a plan that keeps every limit from the best plan.
- Find the best plan for our class outings, and ask who it leaves out.

Activity: The class outings

1. Warm-up (5 min): Ask pupils what would make the best day out. Collect answers (cheap, fun, close, everyone together) and point out that people mean different things by “best”.
2. Teach (8 min): Write three headings on the board: options (what we choose), limits (what must hold) and goal (what “best” means). Fill them in for the outings: five ideas, a budget of 1,200 coins, the most votes.
3. Vote (5 min): Write five outing ideas with their costs, such as zoo 700, museum 500, farm 400, river park 300 and science fair 200. Each pupil votes for two; write the totals beside each idea.
4. Find the best (15 min): Groups use one card per idea and try mixes, adding up the cost of each. A plan over budget is not allowed. Each group writes its best plan and its votes on the board.
5. Wrap-up (7 min): Compare the plans. Did “favourite first” win? Which plan has the most votes? Ask whose favourite was left out and agree one fair rule for next term, such as taking turns.

Prepare one set of five cards per group: the outing, its cost and space to write its votes. No outings planned? Use class rewards instead (a story hour, a football match, a film).

Discussion

- Why didn’t “favourite first” give the most votes?
- How do we know a plan is the best, and not just allowed?
- Whose favourite was left out, and what would be fair next time?

Exit ticket

- Write the options, the limits and the goal of one decision at our school.

Try it in class tomorrow

Plan the class outings together: write five ideas with their costs, let pupils vote, and set a budget. Groups find the mix with the most votes inside the budget and explain why it’s the best. Then ask: whose favourite was left out, and what would be fair next term?

Worksheet · for pupils

What makes a choice the best?

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Name _____ Class _____ Date _____

1 **A colleague says: "The best plan for the class outings always starts with the class favourite, the zoo." Is she right?**

- ◯ No: a mix of smaller choices can score more
- ◯ Only when the favourite is the cheapest
- ◯ Yes: the favourite belongs in every best plan

2 **The school kitchen plans lunch. Match each sentence to the part of the decision it describes.**

How many portions of rice, beans and greens to serve	• • • •	The goal
Every pupil gets at least 5 mg of iron	• • • •	A feasible plan
Spend as little money as possible	• • • •	The options
A plate that meets every need, whatever it costs	• • • •	A limit

3 **Adding a new limit, such as "every class gets at least one outing", can never make the best plan score higher on the goal.**

- ◯ True
- ◯ False

4 **Which of these are limits for a school timetable? Choose all that apply.**

- ☐ As few gaps as possible in teachers' days
- ☐ Which period each lesson goes in
- ☐ Ms Huda leaves after period 3
- ☐ Each class has maths four times a week
- ☐ A teacher can't teach two classes at once

5 **Complete the sentence.**

A plan that keeps every _____ is _____; the one with the best score on the goal is _____.

Word bank: optimal · limit · goal · option · feasible

Answers · for the teacher

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1 **No: a mix of smaller choices can score more**

Favourite first (the zoo and the museum) spends all 1,200 coins for 43 votes.

2 **How many portions of rice, beans and greens to serve — The options**
Every pupil gets at least 5 mg of iron — A limit
Spend as little money as possible — The goal
A plate that meets every need, whatever it costs — A feasible plan

The portions are what the kitchen decides; the iron rule must hold for every plan; the cost is the one number to make small.

3 **True**

A new limit only removes plans; it never adds one.

4 **A teacher can't teach two classes at once; Ms Huda leaves after period 3; Each class has maths four times a week**

Limits must hold in every plan: no teacher in two places, Ms Huda's time, four maths lessons.

5 **A plan that keeps every limit is feasible; the one with the best score on the goal is optimal.**

Feasible means allowed: every limit is kept.

For you to think about

Think of a decision people argue about at your school: the timetable, duties, a trip. What are its options, limits and goal, and do people agree on the goal?